Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: De Carvalho Bros. Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
8342 Maze BLVD	Modesto	Stanislaus	95358

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1972

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0012-0047-0003-0000

B. OPERATORS

Carvalho, Tony

Operator name: Carvalho, Tony
Telephone no.: (209) 548-9317 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
8342 Maze BLVD	Modesto	CA	95358

This operator is responsible for paying permit fees.

De Carvalho, Arturo

Operator name: De Carvalho, Arturo
Telephone no.: (209) 548-9317 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
8342 Maze BLVD	Modesto	CA	95358

De Carvalho, Joe

Operator name: De Carvalho, Joe
Telephone no.: (209) 548-9317 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
8342 Maze BLVD	Modesto	CA	95358

C. OWNERS

Carvalho, Tony

Legal owner name: Carvalho, Tony
Telephone no.: (209) 548-9317
Landline | Cellular

8342 Maze BLVD | Modesto | CA | 95358
Mailing Address Number and Street | City | State | Zip Code

This owner is responsible for paying permit fees.

De Carvalho, Arturo

Legal owner name: De Carvalho, Arturo
Telephone no.: (209) 548-9317
Landline | Cellular

8342 Maze BLVD | Modesto | CA | 95358
Mailing Address Number and Street | City | State | Zip Code

De Carvalho, Joe

Legal owner name: De Carvalho, Joe
Telephone no.: (209) 548-9317
Landline | Cellular

8342 Maze BLVD | Modesto | CA | 95358
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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	60	120	160	65	0
Number under roof	485	0	60	0	0	60
Maximum number	485	60	180	160	65	60
Average number	485	60	180	160	65	60
Avg live weight (lbs)	1,400	1,450	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 92 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 18,407.46 *tons per reporting period*

Total nitrogen from manure: 228,635.54 *lbs per reporting period*

After ammonia losses (30% loss applied): 160,044.88 *lbs per reporting period*

Total phosphorus from manure: 37,905.22 *lbs per reporting period*

Total potassium from manure: 99,892.16 *lbs per reporting period*

Total salt from manure: 242,159.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 11,510,200 *gallons*

Total nitrogen generated: 52,088.79 *lbs*

Total phosphorus generated: 8,643.29 *lbs*

Total potassium generated: 94,015.99 *lbs*

Total salt generated: 654,089.18 *lbs*

	11,415,200	*gallons applied*
+	95,000	*gallons exported*
-	0	*gallons imported*
=	11,510,200	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Modesto Irrigation District	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
10/20/2023	Process wastewater sludge	95,000.00 *gal*	2,740.00	477.00	0.00	0.00	803.00	1,230.00		4,990

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	2,172.20	636.60	975.11	3,955.95
Total exports for all materials	2,172.20	636.60	975.11	3,955.95

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
F1-Dairy	27	27	2	process wastewater	X012-X047-X003-XXXX
F2-Trailer	38	38	2	process wastewater	X012-X047-X004-XXXX
F3-Russell	40	40	2	both	X012-X047-X006-XXXX
F4-Faust	40	40	2	both	X012-X050-X001-XXXX
Totals for areas that were used for application	145	145	8		
Totals for areas that were not used for application					
Land application area totals	145	145	8		

B. CROPS AND HARVESTS

F1-Dairy

Field name: F1-Dairy

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	438.00 *ton*	Dry-weight		63.4	17,700.00	3,900.00	31,400.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	16.22	210.18	46.31	372.86	1,377.46

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	655.00 *ton*	Dry-weight		69.3	15,000.00	3,300.00	29,500.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	1,250.00
Total actual harvest content	24.26	223.43	49.15	439.41	1,668.26

F2-Trailer

Field name: F2-Trailer

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	616.00 *ton*	Dry-weight		65.5	14,100.00	3,400.00	24,500.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	16.21	157.71	38.03	274.04	1,208.01

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	922.00 *ton*	Dry-weight		65.8	9,200.00	2,600.00	13,500.00		5.44

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	1,250.00
Total actual harvest content	24.26	152.68	43.15	224.05	902.82

F3-Russell

Field name: F3-Russell

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	667.00 *ton*	Dry-weight		63.1	20,300.00	3,800.00	28,800.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	16.68	249.81	46.76	354.42	1,230.62

F3-Russell

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	978.00 *ton*	Dry-weight		65.6	8,800.00	3,200.00	9,600.00		5.39

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	1,250.00
Total actual harvest content	24.45	148.03	53.83	161.49	906.68

F4-Faust

Field name: F4-Faust

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	642.00 *ton*	Dry-weight		65.4	24,500.00	4,600.00	43,200.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	1,600.00
Total actual harvest content	16.05	272.11	51.09	479.81	1,266.15

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	524.00 *ton*	Dry-weight		70.6	13,300.00	2,400.00	23,700.00		9.38

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	1,250.00
Total actual harvest content	13.10	102.45	18.49	182.56	722.52

NUTRIENT BUDGET

A. LAND APPLICATIONS

F1-Dairy - 11/11/2022: Oats, silage-soft dough

Field name: F1-Dairy

Crop: Oats, silage-soft dough — Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	183.77	53.86	82.49	334.67	217,000.00 *gal*
Application event totals		183.77	53.86	82.49	334.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	48.12	5.38	105.20	847.85	337,000.00 *gal*
Modesto Irrigation District	Surface water	1.13	0.00	0.00	35.90	3,057,000.00 *gal*
Application event totals		49.25	5.38	105.20	883.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	42.41	4.75	92.71	747.21	297,000.00 *gal*
Modesto Irrigation District	Surface water	1.11	0.00	0.00	35.06	2,985,000.00 *gal*
Application event totals		43.52	4.75	92.71	782.27	

F1-Dairy - 06/09/2023: Corn, silage

Field name: F1-Dairy

Crop: Corn, silage — Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

F1-Dairy - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Modesto Irrigation District	Surface water	1.57	0.00	0.00	49.81	4,241,000.00 *gal*
Application event totals		1.57	0.00	0.00	49.81	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Modesto Irrigation District	Surface water	0.90	0.00	0.00	28.46	2,423,000.00 *gal*
Application event totals		0.90	0.00	0.00	28.46	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	69.32	9.57	202.18	1,569.31	623,000.00 *gal*
Modesto Irrigation District	Surface water	0.90	0.00	0.00	28.46	2,423,000.00 *gal*
Application event totals		70.22	9.57	202.18	1,597.77	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	74.07	8.71	141.54	842.17	521,000.00 *gal*
Modesto Irrigation District	Surface water	0.90	0.00	0.00	28.46	2,423,000.00 *gal*
Application event totals		74.97	8.71	141.54	870.63	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	84.45	9.93	161.38	960.18	594,000.00 *gal*
Modesto Irrigation District	Surface water	0.90	0.00	0.00	28.46	2,423,000.00 *gal*
Application event totals		85.35	9.93	161.38	988.63	

F1-Dairy - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	50.61	5.95	96.72	575.46	356,000.00 *gal*
Modesto Irrigation District	Surface water	0.90	0.00	0.00	28.46	2,423,000.00 *gal*
Application event totals		51.51	5.95	96.72	603.92	

F2-Trailer - 11/11/2022: Oats, silage-soft dough

Field name: F2-Trailer

Crop: Oats, silage-soft dough Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	157.05	46.03	70.50	286.01	261,000.00 *gal*
Application event totals		157.05	46.03	70.50	286.01	

03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	19.99	2.24	43.69	352.15	197,000.00 *gal*
Modesto Irrigation District	Surface water	0.88	0.00	0.00	27.71	3,321,000.00 *gal*
Application event totals		20.86	2.24	43.69	379.87	

04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	22.44	2.51	49.06	395.41	221,200.00 *gal*
Modesto Irrigation District	Surface water	0.92	0.00	0.00	29.14	3,492,000.00 *gal*
Application event totals		23.36	2.51	49.06	424.55	

F2-Trailer - 06/09/2023: Corn, silage

F2-Trailer - 06/09/2023: Corn, silage

Field name: F2-Trailer

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Modesto Irrigation District	Surface water	1.60	0.00	0.00	50.55	6,058,000.00 *gal*
Application event totals		1.60	0.00	0.00	50.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Modesto Irrigation District	Surface water	0.80	0.00	0.00	25.28	3,029,000.00 *gal*
Application event totals		0.80	0.00	0.00	25.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	64.45	7.58	123.16	732.77	638,000.00 *gal*
Modesto Irrigation District	Surface water	0.85	0.00	0.00	26.96	3,231,000.00 *gal*
Application event totals		65.30	7.58	123.16	759.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	60.41	7.10	115.43	686.82	598,000.00 *gal*
Modesto Irrigation District	Surface water	0.74	0.00	0.00	23.59	2,827,000.00 *gal*
Application event totals		61.15	7.10	115.43	710.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	23.88	3.30	69.64	540.51	302,000.00 *gal*
Modesto Irrigation District	Surface water	0.74	0.00	0.00	23.59	2,827,000.00 *gal*
Application event totals		24.62	3.30	69.64	564.11	

F2-Trailer - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	48.08	5.66	91.88	546.70	476,000.00 *gal*
Modesto Irrigation District	Surface water	0.64	0.00	0.00	20.22	2,423,000.00 *gal*
Application event totals		48.72	5.66	91.88	566.92	

F3-Russell - 11/11/2022: Oats, silage-soft dough

Field name: F3-Russell

Crop: Oats, silage-soft dough

Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	157.76	68.79	255.90	458.60	237.00 *ton*
Application event totals		157.76	68.79	255.90	458.60	

03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	41.93	4.69	91.66	738.72	435,000.00 *gal*
Modesto Irrigation District	Surface water	1.13	0.00	0.00	35.83	4,520,000.00 *gal*
Application event totals		43.06	4.69	91.66	774.55	

04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	45.51	6.28	132.75	1,030.38	606,000.00 *gal*
Modesto Irrigation District	Surface water	1.13	0.00	0.00	35.64	4,496,000.00 *gal*
Application event totals		46.64	6.28	132.75	1,066.02	

F3-Russell - 06/09/2023: Corn, silage

F3-Russell - 06/09/2023: Corn, silage

Field name: F3-Russell

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Modesto Irrigation District	Surface water	2.33	0.00	0.00	73.65	9,290,000.00 *gal*
Application event totals		2.33	0.00	0.00	73.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Modesto Irrigation District	Surface water	1.31	0.00	0.00	41.63	5,250,960.00 *gal*
Application event totals		1.31	0.00	0.00	41.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	70.97	9.80	207.01	1,606.78	945,000.00 *gal*
Modesto Irrigation District	Surface water	1.31	0.00	0.00	41.63	5,251,000.00 *gal*
Application event totals		72.29	9.80	207.01	1,648.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	54.51	6.41	104.16	619.75	568,000.00 *gal*
Modesto Irrigation District	Surface water	1.26	0.00	0.00	40.03	5,049,000.00 *gal*
Application event totals		55.77	6.41	104.16	659.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	56.38	7.71	98.31	623.69	563,000.00 *gal*
Modesto Irrigation District	Surface water	1.31	0.00	0.00	41.63	5,251,000.00 *gal*
Application event totals		57.69	7.71	98.31	665.32	

F4-Faust - 11/11/2022: Oats, silage-soft dough

F4-Faust - 11/11/2022: Oats, silage-soft dough

Field name: F4-Faust

Crop: Oats, silage-soft dough

Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Separator solids	164.41	71.69	266.69	477.95	247.00 *ton*
Application event totals		164.41	71.69	266.69	477.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	56.67	6.34	123.90	998.55	588,000.00 *gal*
Modesto Irrigation District	Surface water	1.00	0.00	0.00	31.81	4,012,000.00 *gal*
Application event totals		57.68	6.34	123.90	1,030.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	42.12	4.71	92.08	742.12	437,000.00 *gal*
Modesto Irrigation District	Surface water	1.00	0.00	0.00	31.60	3,986,000.00 *gal*
Application event totals		43.12	4.71	92.08	773.72	

F4-Faust - 06/09/2023: Corn, silage

Field name: F4-Faust

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	71.42	9.86	208.32	1,616.98	951,000.00 *gal*
Modesto Irrigation District	Surface water	1.42	0.00	0.00	44.83	5,655,000.00 *gal*
Application event totals		72.84	9.86	208.32	1,661.81	

F4-Faust - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Modesto Irrigation District	Surface water	1.11	0.00	0.00	35.22	4,443,120.00 *gal*
Application event totals		1.11	0.00	0.00	35.22	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	24.63	3.40	71.85	557.70	328,000.00 *gal*
Modesto Irrigation District	Surface water	0.74	0.00	0.00	23.47	2,961,000.00 *gal*
Application event totals		25.38	3.40	71.85	581.17	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Modesto Irrigation District	Surface water	0.71	0.00	0.00	22.41	2,827,000.00 *gal*
Application event totals		0.71	0.00	0.00	22.41	

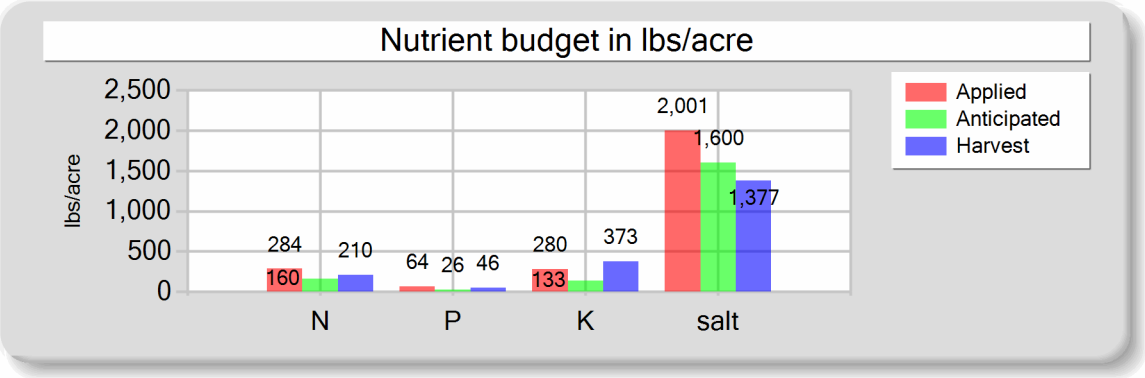
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	34.16	4.02	65.28	388.43	356,000.00 *gal*
Modesto Irrigation District	Surface water	0.71	0.00	0.00	22.41	2,827,000.00 *gal*
Application event totals		34.87	4.02	65.28	410.85	

B. NUTRIENT BUDGET

F1-Dairy - 11/11/2022: Oats, silage-soft dough

Field name: F1-Dairy Crop: Oats, silage-soft dough Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	274.30	63.99	280.41	1,929.73
Fresh water	2.24	0.00	0.00	70.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	283.54	63.99	280.41	2,000.69
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	210.18	46.31	372.86	1,377.46
Nutrient balance	73.36	17.68	-92.46	623.23
Applied to removed ratio	1.35	1.38	0.75	1.45

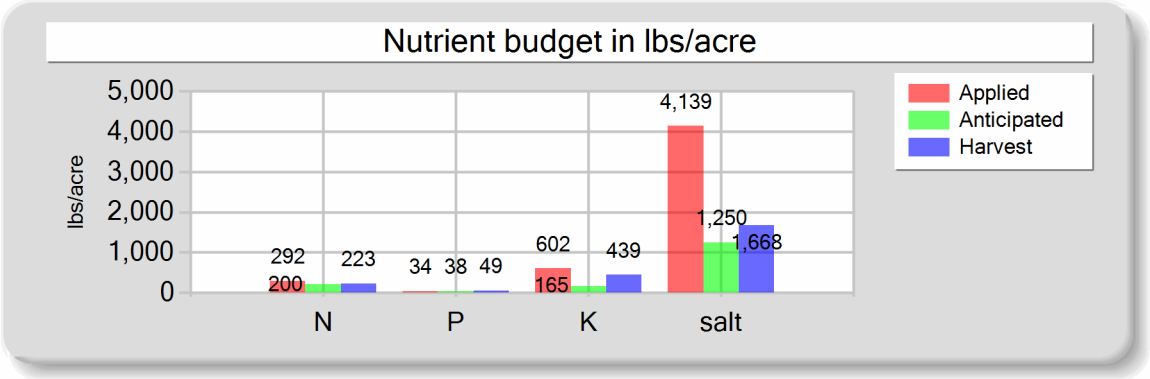
Fresh water applied	
6,042,000.00	*gallons*
222.51	*acre-inches*
8.24	*inches/acre*

Process wastewater applied	
851,000.00	*gallons*
31.34	*acre-inches*
1.16	*inches/acre*

Total harvests for the crop	
1	*harvests*

F1-Dairy - 06/09/2023: Corn, silage

Field name: F1-Dairy Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	278.46	34.17	601.82	3,947.12
Fresh water	6.07	0.00	0.00	192.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.52	34.17	601.82	4,139.22
Anticipated crop nutrient removal	200.00	37.50	165.00	1,250.00
Actual crop nutrient removal	223.43	49.15	439.41	1,668.26
Nutrient balance	68.10	-14.99	162.41	2,470.95
Applied to removed ratio	1.30	0.70	1.37	2.48

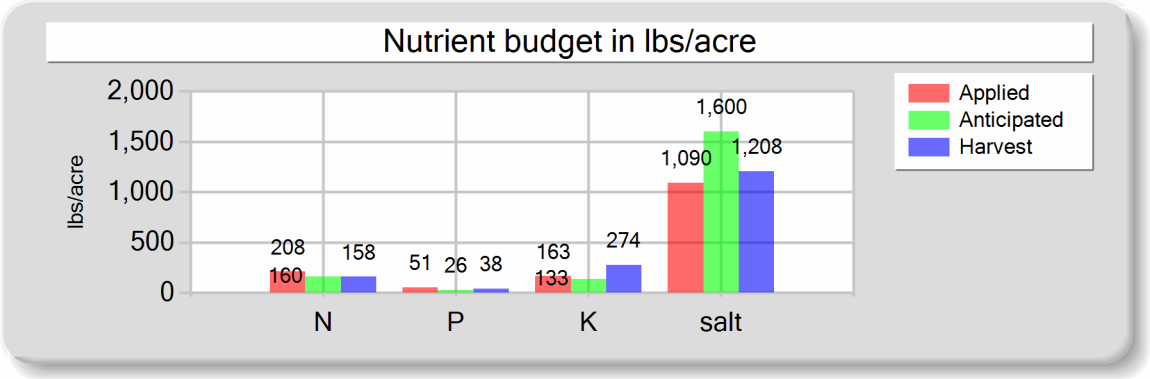
Fresh water applied	
16,356,000.00	*gallons*
602.34	*acre-inches*
22.31	*inches/acre*

Process wastewater applied	
2,094,000.00	*gallons*
77.11	*acre-inches*
2.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

F2-Trailer - 11/11/2022: Oats, silage-soft dough

Field name: F2-Trailer | Crop: Oats, silage-soft dough | Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	199.48	50.77	163.26	1,033.58
Fresh water	1.80	0.00	0.00	56.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.27	50.77	163.26	1,090.44
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	157.71	38.03	274.04	1,208.01
Nutrient balance	50.56	12.74	-110.78	-117.57
Applied to removed ratio	1.32	1.34	0.60	0.90

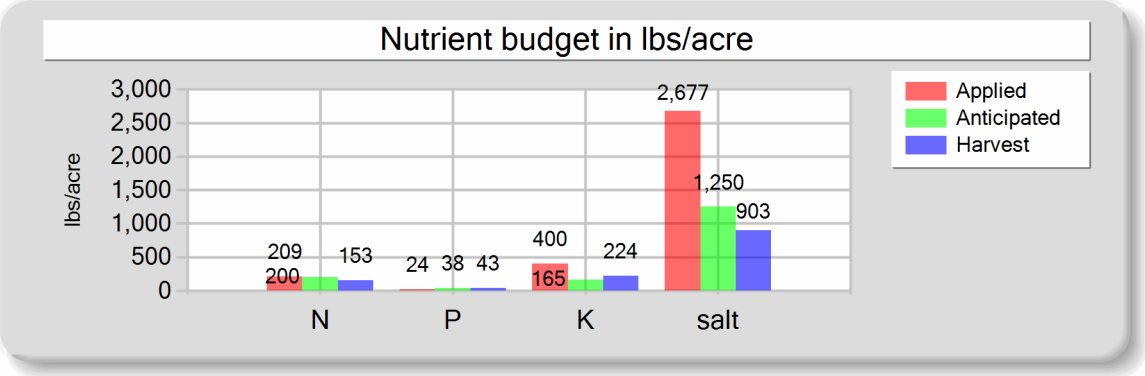
Fresh water applied	
6,813,000.00	*gallons*
250.90	*acre-inches*
6.60	*inches/acre*

Process wastewater applied	
679,200.00	*gallons*
25.01	*acre-inches*
0.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

F2-Trailer - 06/09/2023: Corn, silage

Field name: F2-Trailer Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	196.82	23.64	400.11	2,506.81
Fresh water	5.37	0.00	0.00	170.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.19	23.64	400.11	2,677.00
Anticipated crop nutrient removal	200.00	37.50	165.00	1,250.00
Actual crop nutrient removal	152.68	43.15	224.05	902.82
Nutrient balance	56.51	-19.51	176.06	1,774.18
Applied to removed ratio	1.37	0.55	1.79	2.97

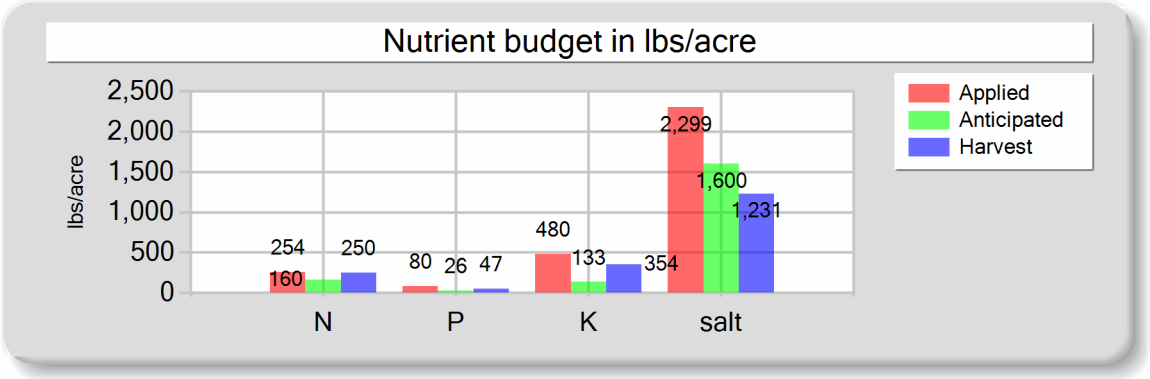
Fresh water applied	
20,395,000.00	*gallons*
751.08	*acre-inches*
19.77	*inches/acre*

Process wastewater applied	
2,014,000.00	*gallons*
74.17	*acre-inches*
1.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

F3-Russell - 11/11/2022: Oats, silage-soft dough

Field name: F3-Russell | Crop: Oats, silage-soft dough | Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	157.76	68.79	255.90	458.60
Process wastewater	87.44	10.98	224.41	1,769.10
Fresh water	2.26	0.00	0.00	71.48
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	254.45	79.76	480.30	2,299.17
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	249.81	46.76	354.42	1,230.62
Nutrient balance	4.64	33.00	125.89	1,068.55
Applied to removed ratio	1.02	1.71	1.36	1.87

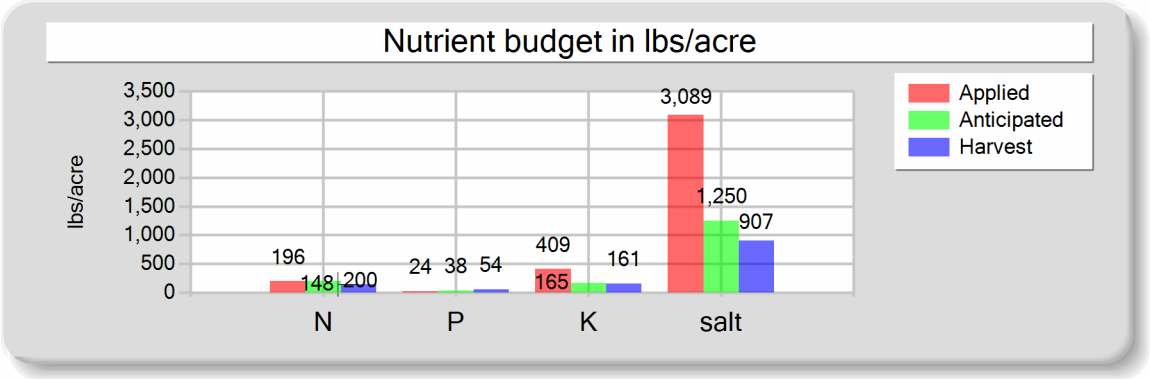
Fresh water applied	
9,016,000.00	*gallons*
332.03	*acre-inches*
8.30	*inches/acre*

Process wastewater applied	
1,041,000.00	*gallons*
38.34	*acre-inches*
0.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

F3-Russell - 06/09/2023: Corn, silage

Field name: F3-Russell | Crop: Corn, silage | Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	181.86	23.91	409.48	2,850.22
Fresh water	7.53	0.00	0.00	238.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.40	23.91	409.48	3,088.78
Anticipated crop nutrient removal	200.00	37.50	165.00	1,250.00
Actual crop nutrient removal	148.03	53.83	161.49	906.68
Nutrient balance	48.37	-29.91	247.99	2,182.10
Applied to removed ratio	1.33	0.44	2.54	3.41

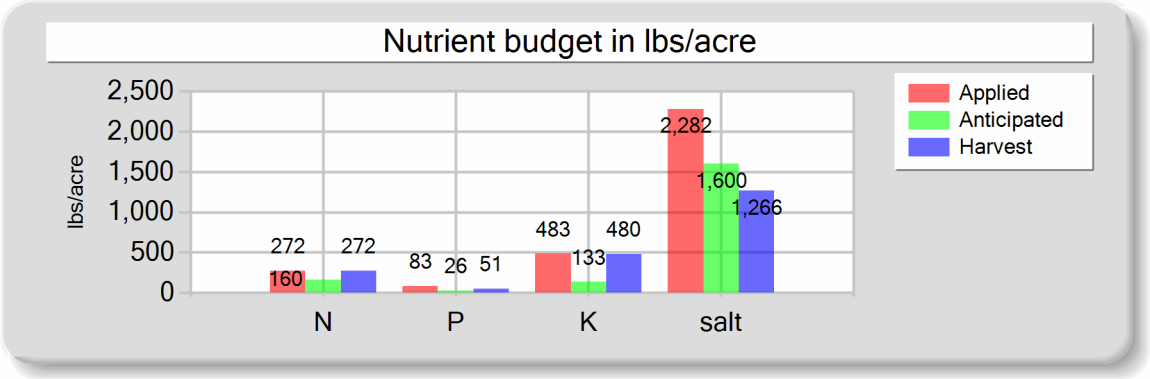
Fresh water applied	
30,091,960.00	*gallons*
1,108.18	*acre-inches*
27.70	*inches/acre*

Process wastewater applied	
2,076,000.00	*gallons*
76.45	*acre-inches*
1.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

F4-Faust - 11/11/2022: Oats, silage-soft dough

Field name: F4-Faust | Crop: Oats, silage-soft dough | Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164.41	71.69	266.69	477.95
Process wastewater	98.79	11.06	215.98	1,740.66
Fresh water	2.00	0.00	0.00	63.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	272.21	82.75	482.67	2,282.01
Anticipated crop nutrient removal	160.00	25.60	132.80	1,600.00
Actual crop nutrient removal	272.11	51.09	479.81	1,266.15
Nutrient balance	0.10	31.66	2.87	1,015.86
Applied to removed ratio	1.00	1.62	1.01	1.80

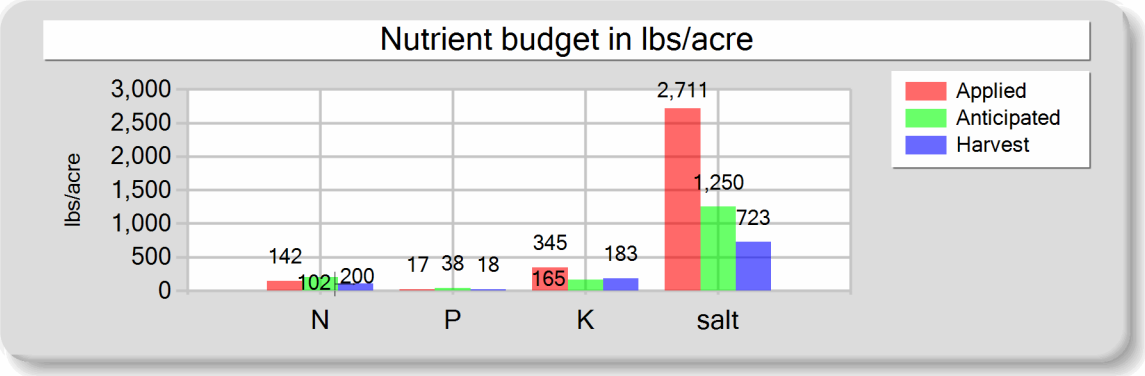
Fresh water applied	
7,998,000.00	*gallons*
294.54	*acre-inches*
7.36	*inches/acre*

Process wastewater applied	
1,025,000.00	*gallons*
37.75	*acre-inches*
0.94	*inches/acre*

Total harvests for the crop	
1	*harvests*

F4-Faust - 06/09/2023: Corn, silage

Field name: F4-Faust Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	130.22	17.28	345.46	2,563.11
Fresh water	4.68	0.00	0.00	148.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	141.91	17.28	345.46	2,711.46
Anticipated crop nutrient removal	200.00	37.50	165.00	1,250.00
Actual crop nutrient removal	102.45	18.49	182.56	722.52
Nutrient balance	39.46	-1.21	162.90	1,988.94
Applied to removed ratio	1.39	0.93	1.89	3.75

Fresh water applied	
18,713,120.00	*gallons*
689.14	*acre-inches*
17.23	*inches/acre*

Process wastewater applied	
1,635,000.00	*gallons*
60.21	*acre-inches*
1.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

M74523-01

Sample and source description: M74523-01

Sample date: 12/21/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 17.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,000.00	6,300.00	22,300.00							5.00
DL	100.00	100.00	100.00							0.10

M74523-02

Sample and source description: M74523-02

Sample date: 12/21/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 22.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	7,500.00	27,900.00							5.00
DL	100.00	100.00	100.00							0.10

B. PROCESS WASTEWATER ANALYSES

L55918-01

Sample and source description: L55918-01

Sample date: 03/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	462.00	444.00	0.00	0.00	51.70	1,010.00								12,300.00	8,140
DL	0.10	0.10	0.10	0.10	0.10	0.10								20.00	12

L55918-02

Sample and source description: L55918-02

Sample date: 06/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	360.00	315.00	0.00	0.00	49.70	1,050.00								12,300.00	8,150
DL	0.10	0.10	0.10	0.10	0.10	0.10								100.00	10

L67240-01

Sample and source description: L67240-01

Sample date: 09/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	460.00	454.00	0.00	0.00	54.10	879.00								7,880.00	5,230
DL	0.10	0.10	0.10	0.10	0.10	0.10								100.00	10

L67240-02

Sample and source description: L67240-02

Sample date: 09/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	480.00	471.00	0.00	0.00	65.60	837.00								7,990.00	5,310
DL	0.10	0.10	0.10	0.10	0.10	0.10								100.00	10

L71187-01

Sample and source description: L71187-01

Sample date: 10/30/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,740.00	477.00	0.00	0.00	803.00	1,230.00								7,520.00	4,990
DL	0.10	0.10	0.10	0.10	0.10	0.10								100.00	10

C. FRESH WATER ANALYSES

Modesto Irrigation District

MID info

Sample description: MID info

Sample date: 07/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.20	0.00	1.20								63.00	38
DL	0.10	0.10	0.10								20.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

F1-Dairy - 11/11/2022: Oats, silage-soft dough

H51816-03

Sample and source description: H51816-03

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	3,900.00	31,400.00		11.60
DL	500.00	200.00	200.00		0.05

F1-Dairy - 06/09/2023: Corn, silage

F1-Dairy - 06/09/2023: Corn, silage

H68873-04

Sample and source description: H68873-04

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	3,300.00	29,500.00		11.20
DL	500.00	200.00	200.00		0.05

F2-Trailer - 11/11/2022: Oats, silage-soft dough

H51816-02

Sample and source description: H51816-02

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	3,400.00	24,500.00		10.80
DL	500.00	200.00	200.00		0.05

F2-Trailer - 06/09/2023: Corn, silage

H68873-02

Sample and source description: H68873-02

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,200.00	2,600.00	13,500.00		5.44
DL	500.00	200.00	200.00		0.05

F3-Russell - 11/11/2022: Oats, silage-soft dough

H51816-04

Sample and source description: H51816-04

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,300.00	3,800.00	28,800.00		10.00
DL	500.00	200.00	200.00		0.05

F3-Russell - 06/09/2023: Corn, silage

H68873-03

Sample and source description: H68873-03

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,800.00	3,200.00	9,600.00		5.39
DL	500.00	200.00	200.00		0.05

F4-Faust - 11/11/2022: Oats, silage-soft dough

H51816-01

Sample and source description: H51816-01

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,500.00	4,600.00	43,200.00		11.40
DL	500.00	200.00	200.00		0.05

F4-Faust - 06/09/2023: Corn, silage

H68873-01

Sample and source description: H68873-01

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	2,400.00	23,700.00		9.38
DL	500.00	200.00	200.00		0.05

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

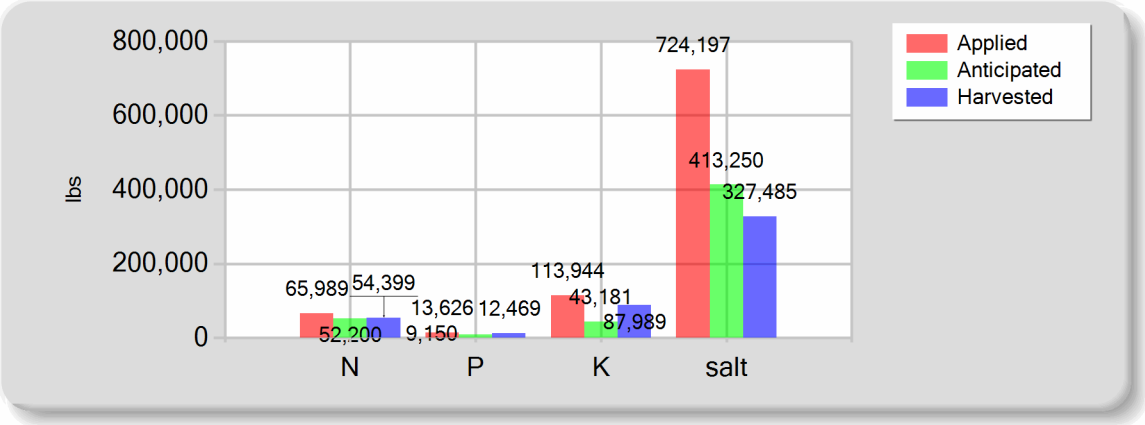
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

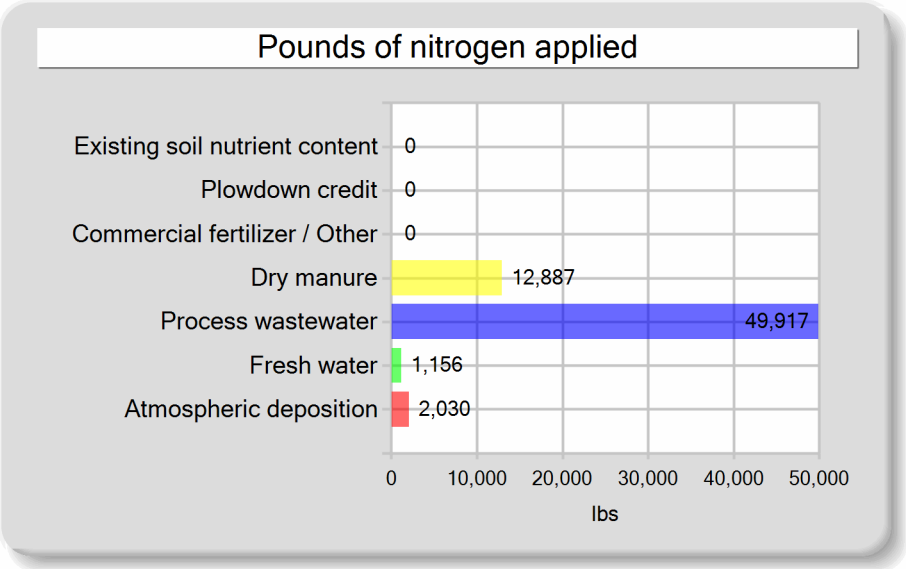
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

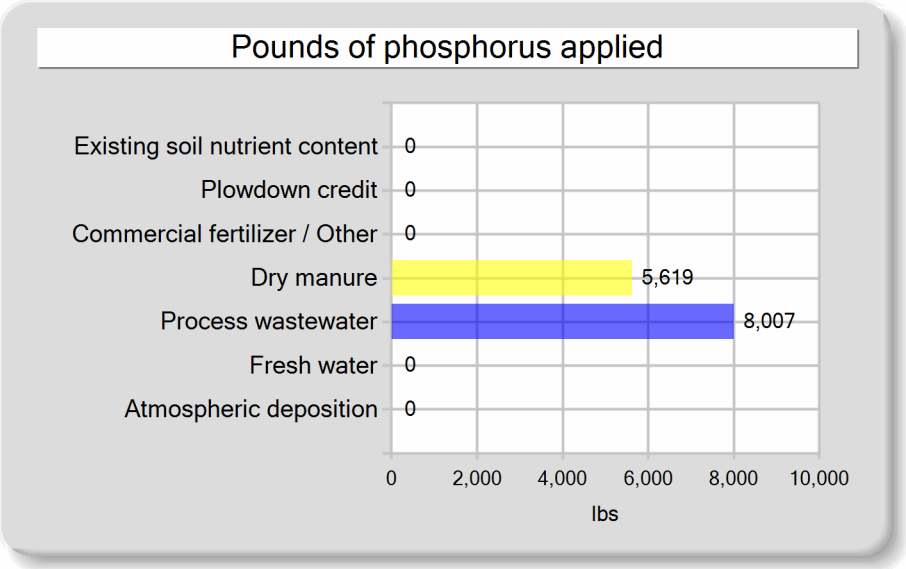
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	12,886.79	5,619.24	20,903.57	37,461.60
Process wastewater	49,916.59	8,006.69	93,040.88	650,133.23
Fresh water	1,155.87	0.00	0.00	36,602.45
Atmospheric deposition	2,030.00	0.00	0.00	0.00
Total nutrients applied	65,989.25	13,625.93	113,944.45	724,197.28
Anticipated crop nutrient removal	52,200.00	9,149.50	43,181.00	413,250.00
Actual crop nutrient removal	54,398.63	12,469.17	87,989.22	327,485.04
Nutrient balance	11,590.61	1,156.76	25,955.22	396,712.24
Applied to removed ratio	1.21	1.09	1.29	2.21

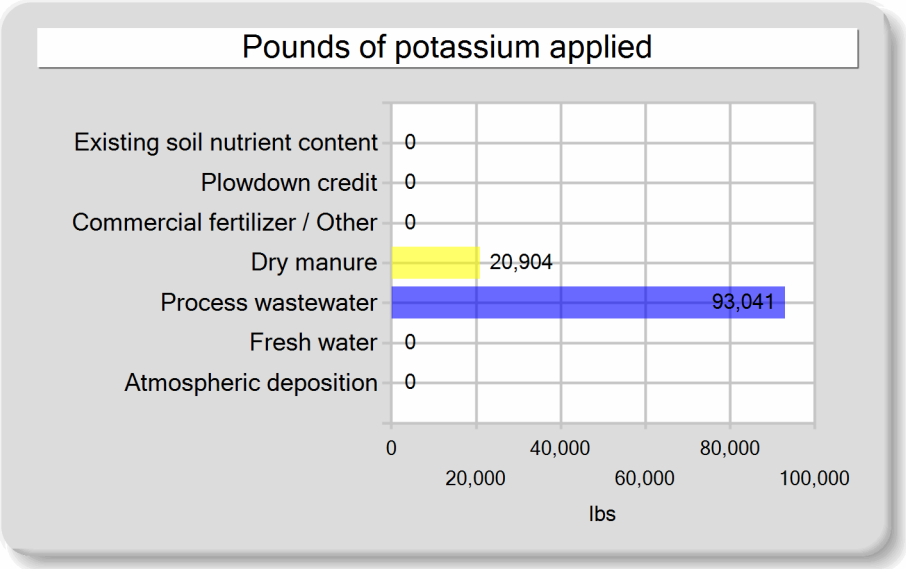
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

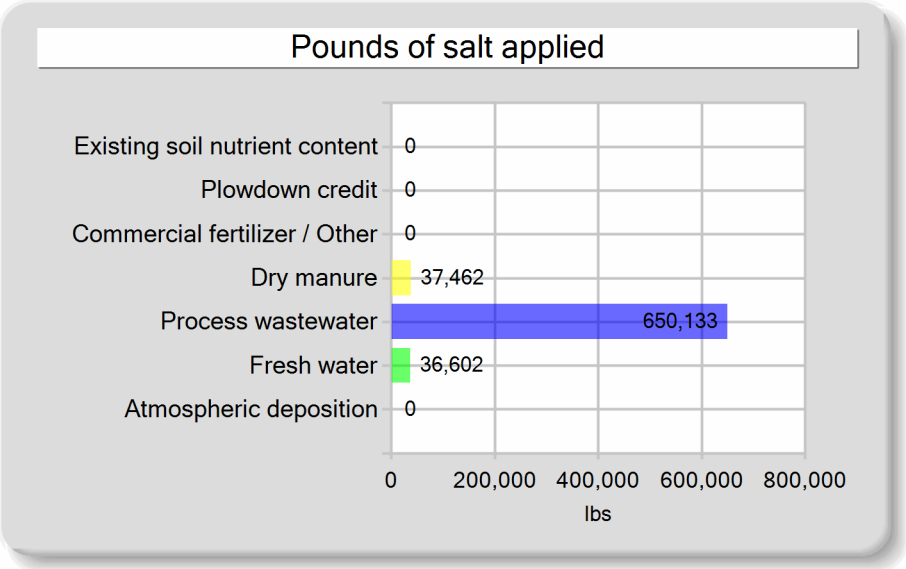


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Tony Carvalho

SIGNATURE OF OWNER OF FACILITY

SIGNATURE OF OPERATOR OF FACILITY

Tony Carvalho

PRINT OR TYPE NAME

SAME AS OWNER

PRINT OR TYPE NAME

6/29/24

DATE

DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

IEH - JL ANALYTICAL

217 Primo Way ▪ Modesto, California 95358 ▪ Office (209) 538-8111 ▪ FAX (209) 538-3966

De Carvalho Bros. Dairy

8342 Maze Blvd.
Modesto,CA 95358

Report #	L2.2-14R66240
Report Date:	01/02/2024
Received Date:	12/21/2023
Work Order:	623417

Laboratory Number: 62341701
Description: Water, #1, Well, Routine
Sampled By:Tony Carvalho, Employed By:DeCarvalho Bros.

Analytical Results

Constituent	Result		Reporting Limit	Method Detection Limit	Method Reference	Analysis Date	Analysis Time	Analyst
Ammoniacal Nitrogen by Kjeldahl	< 2	mg/l	2	0.3	SM 4500 NH3 C	12/26/2023	08:00	JN
Electrical Conductivity (E.C.)	1100	umho/cm	1	1	SM 2510B[5]	12/29/2023	12:05	KB
Nitrate Nitrogen	17	mg/l	0.1	0.01	EPA 300.0	12/22/2023	09:05	JDLC

QC Results

Constituent	QC Units	Blank	Matrix	Theoretical Spike	Matrix Spike	Matrix Spike Duplicate	Lab Control Spike	Precision %	Accuracy %	Response %
Ammoniacal Nitrogen by Kjeldahl	mg/l	< 2	< 2	28	28	28	28	1.5	99.4	98.2
Electrical Conductivity (E.C.)	umho/cm	< 1	73	1400	1500	1500	1400	0.3	100.7	100.1
Nitrate Nitrogen	mg/l	< 0.1	8.6	0.9	9.5	9.4	0.9	0.4	95.6	97.3

Notes

5 At 25 °C

Authorized By: Britney Dundee 01/02/2024

Britney Dundee, Chemistry Supervisor

ELAP Accreditation Laboratory Certificate #2776. Unless otherwise noted, all samples were received in acceptable condition. The result(s) in this report relate only to the portion of the sample(s) tested. Results are not corrected for recovery or expanded measurment of uncertainty. This report does not constitute a release of product for consumption. This report shall not be reproduced except in full, without written approval of the laboratory. This document contains confidential commercial information pursuant to 5 U.S.C. SEC. 552(b)(4).

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De Carvalho Bros. Dairy

8342 Maze Blvd.
Modesto,CA 95358

Report #	L2.2-14R66240
Report Date:	01/02/2024
Received Date:	12/21/2023
Work Order:	623417

Laboratory Number: 62341702
Description: Water, #2, Well, Routine
Sampled By:Tony Carvalho, Employed By:DeCarvalho Bros.

Analytical Results

Constituent	Result		Reporting Limit	Method Detection Limit	Method Reference	Analysis Date	Analysis Time	Analyst
Ammoniacal Nitrogen by Kjeldahl	< 2	mg/l	2	0.3	SM 4500 NH3 C	12/26/2023	08:00	JN
Electrical Conductivity (E.C.)	1100	umho/cm	1	1	SM 2510B[5]	12/29/2023	12:05	KB
Nitrate Nitrogen	17	mg/l	0.1	0.01	EPA 300.0	12/22/2023	09:05	JDLC

QC Results

Constituent	QC Units	Blank	Matrix	Theoretical Spike	Matrix Spike	Matrix Spike Duplicate	Lab Control Spike	Precision %	Accuracy %	Response %
Ammoniacal Nitrogen by Kjeldahl	mg/l	< 2	< 2	28	28	28	28	1.5	99.4	98.2
Electrical Conductivity (E.C.)	umho/cm	< 1	73	1400	1500	1500	1400	0.3	100.7	100.1
Nitrate Nitrogen	mg/l	< 0.1	8.6	0.9	9.5	9.4	0.9	0.4	95.6	97.3

Notes

5 At 25 °C

Authorized By: Britney Dundee 01/02/2024

Britney Dundee, Chemistry Supervisor

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De Carvalho Bros. Dairy

8342 Maze Blvd.
Modesto,CA 95358

Report #	L2.2-14R66240
Report Date:	01/02/2024
Received Date:	12/21/2023
Work Order:	623417

Laboratory Number: 62341703
Description: Water, #3, Well, Routine
Sampled By:Tony Carvalho, Employed By:DeCarvalho Bros.

Analytical Results

Constituent	Result		Reporting Limit	Method Detection Limit	Method Reference	Analysis Date	Analysis Time	Analyst
Ammoniacal Nitrogen by Kjeldahl	< 2	mg/l	2	0.3	SM 4500 NH3 C	12/26/2023	08:00	JN
Electrical Conductivity (E.C.)	1700	umho/cm	1	1	SM 2510B[5]	01/02/2024	12:40	JDLC
Nitrate Nitrogen	38	mg/l	0.1	0.01	EPA 300.0	12/22/2023	09:05	JDLC

QC Results

Constituent	QC Units	Blank	Matrix	Theoretical Spike	Matrix Spike	Matrix Spike Duplicate	Lab Control Spike	Precision %	Accuracy %	Response %
Ammoniacal Nitrogen by Kjeldahl	mg/l	< 2	< 2	28	28	28	28	1.5	99.4	98.2
Electrical Conductivity (E.C.)	umho/cm	< 1	560	1400	1900	1900	1400	0.4	93.1	100.0
Nitrate Nitrogen	mg/l	< 0.1	8.6	0.9	9.5	9.4	0.9	0.4	95.6	97.3

Notes

5 At 25 °C

Authorized By: Britney Dundee 01/02/2024

Britney Dundee, Chemistry Supervisor

ELAP Accreditation Laboratory Certificate #2776. Unless otherwise noted, all samples were received in acceptable condition. The result(s) in this report relate only to the portion of the sample(s) tested. Results are not corrected for recovery or expanded measurment of uncertainty. This report does not constitute a release of product for consumption. This report shall not be reproduced except in full, without written approval of the laboratory. This document contains confidential commercial information pursuant to 5 U.S.C. SEC. 552(b)(4).

Company Name: DeCarvalho Bros.
Attn: Tony C.
Address: 8392 Maze Blvd. Modesto
Phone: (209) 523-2198
Fax:
Email: tonyc8392@hotmail.com

JL Analytical Services
217 Primo Way
Modesto, CA 95358
Phone (209) 538-8111
FAX (209) 538-3966

2F077 ver. 3 ELAP chain of custody

Analysis Requested (write in the sets)

Lab Use Only: Work Order Number	Matrix	Sample Description	Sampling Date	Sampling Time	Chlorine Level	Field pH	Sample Type	Reason For Analysis	Sample Preserv.	EC	Ammonium Nitrogen	Nitrate - N							Temp as read	Corr. Temp
		Sample Information:								Check the Appropriate Box Below:									Thermometer ID:	
	DW	Well # 1	12/20/23	4:00			W	A		X	X	X								
	DW	Well # 2	"	"			W	A		X	X	X								
	DW	Well # 3	"	"			W	A		X	X	X								

Sampled by: Tony Carvalho				
Relinquished by (signature): Tony Carvalho	Received by (signature): Kelly Wong		Date: 12/21/23	Time: 1206
Relinquished by (signature):	Received by (signature):		Date:	Time:

Key:

Sub sample bottle ID: If more than one bottle per sample assign a unique letter to each (A, B, C, etc.)

Matrix: *DW=Drinking Water* *GW=Ground Water* *SW=Storm Water* *WW=Waste Water*

Sample Type: *D=Distribution* *W=Well* *O=Other*

Reason for Analysis: *A=Routine* *B=Repeat* *C=Special* *D=Other*

Sample Preserv.: *M: Cool, <10 deg C, Sodium Thiosulfate (Na2S2O3),* *C: <= 6 deg C* **NA:** Nitric Acid (HNO3) **SA:** Sulfuric Acid (H2SO4) **H:** Hydrochloric Acid (HCl)

Notes / Special Instructions:

Please forward copy of results to: psousa91@yahoo.com (Dairy Monitoring)

Data/Admin/Middesk/Front Office/Chain of Custody's

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Tony Carvalho

Name of Dairy Facility: De Carvalho Bros. Dairy

Facility Address:

8342 Maze BLVD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Tony Carvalho (209) 523-2198
Name Phone Number

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Avila Spreading Service

Address of Hauling Company/Person:

5249 S Carpenter RD	Modesto	CA	95358
Number and Street	City	State	Zip Code

Contact Person: Anthony Avila (209) 523-0266
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Wheeler Farming (209) 526-9770
Name Phone Number

5301 Woodland AVE	Modesto	CA	95358
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

201 Russell RD	Modesto	95358
Address	City	Zip Code

Stanislaus
Street and nearest cross street (if no address) County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 10/20/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 95,000 gallons

Method used to determine volume of process wastewater:

Load volume X number of loads

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Tony Carvalho — 6/29/24

Operator Signature — Date

Hauler Signature — Date